

CSE 260 LAB REPORT 4

Experiment Name:

Applications of K-map Method

Group No. : 5

Members:-

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Section : 22

Objective:

- To investigate the rules of kmap
- To gain experience working with practical circuits
- To simplify ~~to~~ a complex function using kmap.

Required components and equipments:-

- AT-700 portable analog/digital laboratory
- AND, OR, NOT gate
- wires

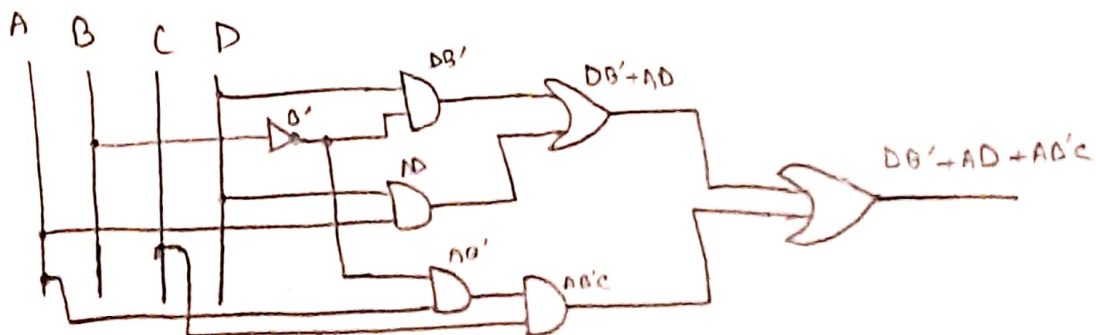
Experimental Setup:

① Circuit diagram for the first function:-

$$F(A, B, C, D) = \Sigma(1, 3, 9, 10, 11, 13, 15)$$

Simplified boolean equation using Kmap:-

$$F = DB' + AD + AB'C$$



Truth Table (function 1)

①

A	B	C	D	Output
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

Kmap

0 - A'
1 - A

	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	1	1	0
10	0	1	1	1

Equ: $DB' + AD + AB'C$

Ant Juf
2.11.22

From the given function, we made a truth table and constructed a K-map. Using the K-map we simplified the ~~boolean~~ equation which came as $DB' + AD + AB'C$. After constructing the circuit, the final output matched with the truth table.

② $F(A, B, C, D) = \sum(1, 4, 15) + d(3, 5, 7, 12, 13, 14)$

Truth table

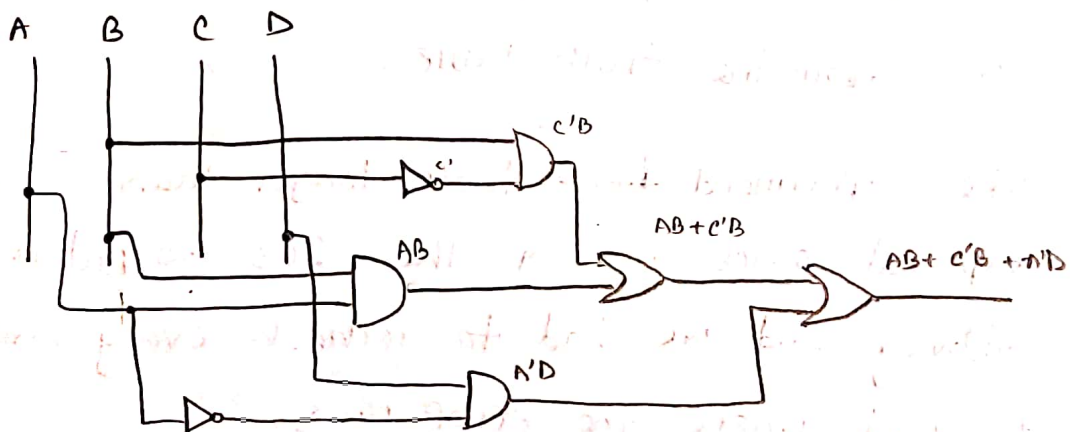
A	B	C	D	output
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

K-map

CD \ AB	c'd'	c'd	cd	cd'
AB		1	X	
A'B'		1	X	
A'B	1	X	X	
AB	X	X	1	X
AB'				

$\therefore F = AB + A'D + c'B$

Circuit diagram:



Discussion:-

For the first function, we drew a Kmap based on the truth table to find the simplified boolean equation.

The equation came as $F = DB' + AD + AB'C$.

Using this equation we constructed a circuit on the breadboard.

We ~~used~~ need four AND gates, ~~one or two~~ OR gates and one NOT gate. So we used one IC for each.

For each IC, pin 14 is the VCC that was connected to the "+5V" position and pin 7 is the ground which is connected to the "GND" position.

The circuit is constructed as shown in the diagram earlier. When tried different combination of the switches, the output matched the one from the truth table.

The experiment ~~loop~~ took longer than expected since one of the IC's stopped working midway and we had to recheck every connection to find where the error was.